EC551 PA2 Work Plan

Stage 1: Compiler front end

Ayla

- [ ] Parser and input validation (Part 1)
- [ ] SSA conversion and `--export-ssa` (Part 2)
- [ ] DAG construction (Part 3)
- [ ] ASAP scheduler (Part 3)

Combine 1: Ayla pushes this code to the shared repo.

- [ ] Fritz can run an equation file through SSA, the DAG, and ASAP
- [ ] The schedule output matches the agreed format

Stage 2: Separate builds

Ayla

- [ ] Constrained ASAP scheduler (Part 3)
- [ ] Adder, subtractor, and multiplier modules (Part 5)
- [ ] FSM generator (Part 4)
- [ ] Testbench generator (Part 6)
- [ ] FPGA top-level module: switch inputs, step button, output selection, and display (Parts 6–7)
- [ ] Constraints file (Part 7)

Fritz

- [ ] ALAP scheduler (Part 3)
- [ ] Constrained ALAP scheduler (Part 3)
- [ ] Custom scheduler (Part 3)
- [ ] Resource binding and sample binding output (Part 4)
- [ ] Mux and register generator (Part 4)
- [ ] Generated system module with functional units and exposed unit outputs (Part 6)
- [ ] One-command build-and-program script (Part 7)

Stage 3: Combine and test

Combine 2: Merge all five schedulers.

- [ ] Run the same test equations through every mode
- [ ] Confirm that dependencies and resource limits are respected

Combine 3: Merge the datapath, FSM, and functional units.

- [ ] Generate the complete Verilog from an equation file
- [ ] Pass `testbench.v` in simulation for several schedules

Combine 4: Add the FPGA top-level module and build flow.

- [ ] Test the board wrapper with one simple design
- [ ] Run the generated design on the FPGA
- [ ] Confirm that stepping, unit selection, and displayed values work